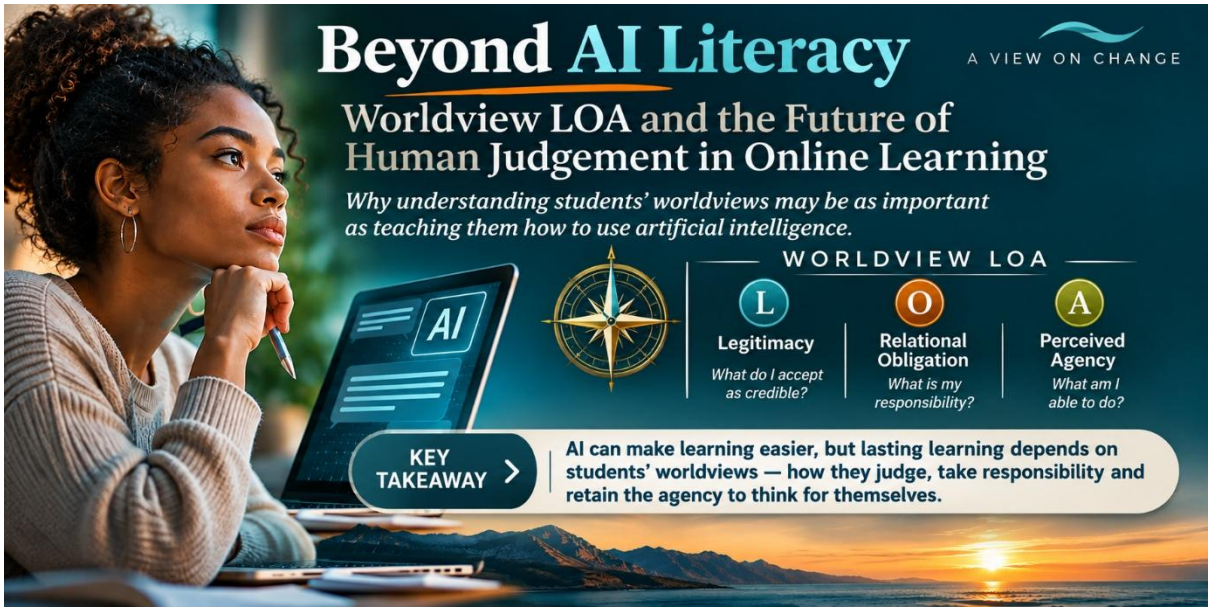


Generative AI can explain complex concepts, summarise academic literature, give feedback, construct arguments and produce written assignments. For students learning online, these capabilities matter a great deal. Much of their study is independent and asynchronous, often with limited direct contact with a lecturer, and AI offers help at any hour.

The educational opportunities are considerable. AI can widen access to information, offer alternative explanations and help students engage with difficult material.

These developments also raise a more fundamental question:

Are we using artificial intelligence to develop students' capacity to think, or are we creating learning environments in which they increasingly delegate their thinking?

This goes beyond academic integrity and plagiarism. It concerns how students decide what counts as credible knowledge, what responsibilities they hold as learners, and whether they believe they can develop understanding for themselves.

These are the concerns of Worldview LOA, a framework that emerged from my doctoral research at the University of Cape Town's Graduate School of Business. It examines three evaluative structures that shape how people judge and act:

- **Legitimacy:** what people accept as credible and appropriate
- **Relational Obligation:** what they believe they owe to others
- **Perceived Agency:** what they believe they are able to do

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Applying the framework to students and AI is an extension of that work, and one that still needs testing. Its value here is that it moves the discussion from how students use AI to the worldviews that shape their engagement with it.

Producing an answer and developing knowledge

Learning has always involved more than acquiring information. Students must learn to interpret, evaluate, question and apply knowledge, and to tell evidence from assertion.

That process is rarely comfortable. A student may misunderstand a concept, doubt an established explanation or struggle to connect theory with practice. Such moments are often part of learning, not evidence of its failure.

Generative AI changes the conditions under which this happens. A student who is stuck can ask AI to explain a concept in simpler language, which can be enormously helpful. The same student can also ask it to interpret the theory, apply it to a case study and draft the answer to an assessment question.

The result may be academically coherent, yet the student may not have done the intellectual work behind it. A completed assignment shows that an acceptable artefact exists. It doesn't show that the student can explain the argument, defend its assumptions or apply the knowledge elsewhere.

So the useful question is whether a student's engagement with AI strengthens or weakens the intellectual processes that higher education exists to develop.

When plausibility passes for knowledge

Quattrociocchi, Capraro and Perc (2025) argue that similar outputs from people and AI can conceal very different processes. A student and an AI system may produce similar explanations without reaching them through equivalent experience, understanding or judgement.

The authors identify seven fault lines between human and artificial judgement: grounding, parsing, experience, motivation, causality, metacognition and value. They also introduce the term *Epistemia*, a condition in which linguistic plausibility substitutes for epistemic evaluation.

In education, this matters because a fluent, persuasive explanation may not be adequately grounded in evidence. A student may accept it because it sounds authoritative, gaining an impression of knowledge without the understanding that should accompany it.

Worldview LOA offers a way to connect these fault lines to student behaviour:

Seven epistemological fault lines: Implications for online learning and Worldview LOA

Epistemological fault line	Implication for online learning	Worldview LOA contribution
1. Grounding	AI works primarily with representations of information, whereas students bring embodied experience and interaction with the world. Students need to connect AI explanations to actual experience and context.	**Legitimacy:** Helps explain how students evaluate the credibility and contextual relevance of AI-generated knowledge, rather than accepting information simply because it is convincingly presented.
2. Parsing	AI processes information through computational representations. Human interpretation is contextual and can be revised as understanding develops. Students need to question how information has been framed and interpreted.	**Legitimacy:** Illuminates the evaluative beliefs through which students accept, question or reject particular interpretations and recognise alternative ways of understanding information.
3. Experience	AI draws on learned representations rather than personal autobiographical experience. Students bring lived experiences, prior learning and situated understanding to academic interpretation.	**Legitimacy and Relational Obligation:** Explains how students draw on their own experiences to evaluate knowledge and connect academic understanding to the circumstances and people affected by its application.
4. Motivation	AI operates through computational objectives, whereas students have personal goals, interests, emotions and motivations. These influence their engagement, persistence and willingness to learn.	**Relational Obligation and Perceived Agency:** Helps explain how students understand their responsibilities as learners, their commitment to learning and their perceived capacity to invest effort in developing their own understanding.

5. Causality	AI can identify patterns and generate plausible explanations, but students must distinguish correlation from causation and interrogate the assumptions behind explanations.	**Perceived Agency:** Provides a lens for understanding whether students actively interrogate causal explanations, recognise their limitations and exercise independent judgement rather than defer to apparently authoritative conclusions.
6. Metacognition	Students can reflect on their understanding, recognise uncertainty and deliberately suspend judgement. They need to retain these capacities when AI provides apparently confident answers.	**Perceived Agency:** Helps explain students' capacity and willingness to recognise the limits of their knowledge, withhold judgement, seek further evidence and take ownership of their learning.
7. Value	AI-generated responses do not carry human moral responsibility in the same way that students and educators do. Students must evaluate the ethical and social implications of knowledge and its application.	**Relational Obligation:** Explains how students recognise responsibilities towards others, consider the consequences of applying knowledge and understand the ethical obligations associated with their academic and future professional roles.

The table summarises the epistemological distinctions for educational application. The LOA contributions are conceptual intersections, not exclusive or deterministic mappings. Each fault line may have implications across more than one dimension of Worldview LOA.

The fault lines describe how human and artificial judgement differ. Worldview LOA asks why those differences affect some learners' behaviour more than others.

Consider two students who both know that AI has no personal experience. One accepts its recommendations uncritically, while the other treats them as the start of an independent investigation. Knowing a technology's limits doesn't automatically produce critical engagement, and the student's evaluative orientation may help explain the difference.

The importance of being able to say "I don't know"

A striking contribution to this debate comes from a recent preprint by Marcoccia, Quattrociocchi and Capraro (2026). Across five experiments involving 3,132 online participants, they asked people about small visual details in films, using an AI model that was almost always wrong.

In one experiment, the share of "I don't know" responses fell from 44% without AI to 3% with it. Without incentives, people answered more questions but were correct only about a third as often. In the experiment that measured confidence, it rose from about 30 to 76 out of 100.

The effect also held when AI advice appeared automatically, without participants asking for it. That finding is relevant to online study, where search summaries and writing assistants increasingly offer answers unprompted.

The study has clear limits. It wasn't conducted specifically with students, and it didn't test what happens when AI is usually right or how the effect develops with repeated use. It does raise an important question for educators: what happens when students become accustomed to having an apparently authoritative answer always at hand?

The willingness to say "I don't know" can be an expression of intellectual maturity. Independent scholarship depends on recognising when evidence is insufficient, asking better questions and accepting that knowledge is provisional.

Within Worldview LOA, this connects most strongly with Perceived Agency. Withholding judgement may partly reflect whether students see themselves as capable of investigating uncertainty, rather than depending on an external source to resolve it.

Worldview or situation?

There is an obvious objection to all of this. If almost everyone stops saying "I don't know" once AI is available, perhaps the situation is doing the work, leaving little room for differences in worldview.

The experiments offer a partial response. When accuracy carried a small financial reward and penalty, people sought AI advice less, answered more accurately and withheld judgement a little more often. The authors conclude that AI changes people's willingness to withhold judgement, not their capacity to do so.

If judgement can be re-engaged when people have a reason to use it, money is unlikely to be the only reason. A student's sense of what counts as credible, what they owe others and what they can achieve may also matter. This is a proposition for Worldview LOA to test, not a finding. The evidence so far suggests that the conditions of learning matter a great deal, so worldview is best understood as working alongside them.

When AI begins to shape worldviews

The relationship may also run in the other direction, with repeated AI use influencing the worldviews students bring to learning.

Consider a student who routinely uses AI to interpret academic content, formulate arguments and resolve uncertainty. At first, this may be a practical response to the demands of online study. Over time, the student may come to expect that explanations will always be available and that uncertainty will be resolved by an external source.

That could affect how they judge information, understand academic responsibility and perceive their own capacity to learn. Two proposed concepts help frame this possibility:

- **Epistemic delegation:** a student transfers part of the interpretive and evaluative work of learning to an AI system.
- **Evaluative adaptation:** repeated engagement with AI gradually shifts the evaluative habits through which a student approaches knowledge, authority and learning.

These are analytical concepts, not established causal findings. Their value lies in the questions they open:

- Does repeated delegation influence perceived agency?
- Does reliance on AI-generated explanations change how students establish legitimacy?
- Under what conditions does AI support independent learning, and when does it encourage dependence?

From AI literacy to worldview-informed learning

Most institutional responses to AI focus on AI literacy, responsible use and academic integrity. These remain necessary, but they leave the deeper evaluative processes of learning largely untouched. Worldview LOA suggests three complementary priorities:

- **Legitimacy: developing epistemic judgement.** Students learn to examine the assumptions behind an explanation, compare AI responses with academic sources and distinguish persuasive presentation from justified knowledge.
- **Relational Obligation: developing intellectual responsibility.** Students take ownership of what they claim to know and consider the consequences of applying that knowledge in professional and social contexts.

- **Perceived Agency: developing independent judgement.** Students retain confidence in their ability to investigate and interpret, using AI as a resource without relying on it for every argument or decision.

These priorities can inform how learning activities and assessments are designed. Students could be asked to:

- evaluate an AI-generated explanation against the academic literature
- identify its unsupported assumptions
- defend their own interpretation
- explain what they still don't understand
- show how their thinking developed, including where they challenged an AI response

None of this requires discouraging AI use. Students can use AI freely while keeping responsibility for their own learning.

A proposal: the Defended Judgement Assessment

I propose an assessment that allows AI but gives most of the marks to the judgement a student exercises around it. It suits a case-based question in an online module and runs in four stages:

Stage	What the student does	What it develops	Suggested weight
1. Commit	Before using AI, writes a short initial position on the case, rates their confidence and lists what they don't yet know.	**Perceived Agency:** forming a view and recognising its limits	10%
2. Interrogate	Puts the question to AI, then annotates the response. Checks its key claims against academic sources and identifies unsupported assumptions.	**Legitimacy:** distinguishing plausible from justified knowledge	30%

3. Take a position	Submits a final argument, explaining what changed and why, who would be affected if it were applied, and what remains uncertain.	**Relational Obligation:** owning a claim and its consequences	30%
4. Defend	In a 10 to 15 minute live online conversation, explains the argument and applies it to a changed version of the case.	**All three:** showing the judgement is the student's own	30%

How to mark it

- **Reward calibration.** A well-founded "I don't know" costs no marks, while a confident claim without support does. The experiments suggest that consequences can shift behaviour, so the marking scheme should make withholding judgement a sensible choice.
- **Mark reasoning, not the verdict on AI.** A student may accept an AI response if they can justify doing so, and may lose marks for rejecting it without good reason.
- **Assess judgement, not worldview.** Worldview LOA informs the design, but no student should be marked on what they believe. The criteria concern evidence, responsibility and independent reasoning.
- **Make the defence decisive.** AI can help draft the written stages, including the reflection. A student who can't explain or adapt their argument in the live conversation shouldn't pass on the written work alone.

Running it online

The live defence is the main practical constraint. Short small-group sessions could help with larger cohorts, although individual conversations offer stronger assurance. All students also need access to a suitable AI tool, and the first stage should carry low stakes because it can't be fully verified online.

This format is a proposal and hasn't been tested. A pilot could compare it with a conventional assignment, looking at whether students become better calibrated, more willing to say "I don't know" and better able to apply their learning in a new context.

Defended Judgement Assessment: marking rubric

The rubric has nine criteria totalling 100 marks. For each one, choose the band that best fits the evidence, then award marks within that band's percentage range. For example, a Proficient result on a 15-mark criterion earns 9 to 11 marks.

Criterion	Excellent (75–100%)	Proficient (60–74%)	Developing (50–59%)	Insufficient (0–49%)
Stage 1: Commit (10 marks)				
1. Initial position and known limits (10)	Takes a clear position, gives reasons, and identifies specific gaps in knowledge that match the stated confidence.	Takes a clear position and identifies relevant gaps, with some reasoning.	Takes a position, but the reasoning or the listed gaps are general.	Gives no real position, or does not identify any gaps.
Stage 2: Interrogate (30 marks)				
2. Checking claims against sources (15)	Tests the most important claims against strong academic sources and shows clearly which hold, which fail and which can't be confirmed.	Checks key claims against suitable sources and reports the results accurately, with minor gaps.	Checks some claims, but misses important ones or relies on weak sources.	Accepts claims without checking, or misreports what sources say.
3. Identifying assumptions and limits (15)	Exposes unstated assumptions and explains how they affect the conclusion in this context.	Identifies the main assumptions and some of their effects.	Identifies assumptions superficially or comments only on	Identifies no meaningful assumptions or limits.

			general AI limitations.	
Stage 3: Take a position (30 marks)				
4. Final argument and development of thinking (15)	Makes a coherent, well-evidenced argument and explains precisely what changed from Stage 1 and why, or why the position held.	Makes a sound, evidenced argument and explains the main changes in thinking.	Makes a basic argument with limited evidence or a descriptive account of what changed.	The argument is unsupported or incoherent, or the development of thinking is not explained.
5. Responsibility and consequences (10)	Identifies who would be affected, weighs the consequences and takes clear ownership of the recommendation.	Identifies the main stakeholders and consequences, and accepts responsibility for the argument.	Mentions stakeholders or consequences with limited analysis.	Ignores consequences or shifts responsibility for the claims to AI or other sources.
6. Calibrated uncertainty (5)	States precisely what remains uncertain, and confidence matches the evidence throughout.	Identifies the main uncertainties, and confidence is largely justified.	Acknowledges uncertainty in general terms, or confidence is sometimes unjustified.	Makes confident claims without support, or hedges so heavily that no position is taken.
Stage 4: Defend (30 marks)				

7. Explanation and ownership (15)	Explains the argument, evidence and assumptions clearly and without relying on the script, and answers probing questions well.	Explains the main argument and evidence accurately, with occasional prompting.	Explains parts of the argument, but needs substantial prompting or struggles with the reasoning.	Can't explain the argument or key evidence in the submitted work.
8. Application to a changed case (10)	Adapts the argument convincingly and explains which parts still apply, which don't and why.	Applies the argument appropriately to the new case, with some explanation.	Makes a partial or mechanical application to the new case.	Can't apply the argument, or repeats the original answer.
9. Judgement under questioning (5)	Recognises the limits of their knowledge, says "I don't know" where appropriate and suggests how to find out.	Acknowledges most limits and responds to challenge without losing the thread.	Sometimes guesses or concedes without reasoning, but can recover when prompted.	Persistently bluffs, or gives way on every challenge without reasoning.

Applying the rubric

- **Defence threshold:** A student must earn at least 15 of the 30 Stage 4 marks to pass. If they earn fewer, cap the overall mark at 49%.
- **Well-founded "I don't know" answers:** Don't deduct marks for these. Where the answer shows sound judgement, credit it under criteria 6 and 9.
- **Position on AI:** Don't reward agreeing or disagreeing with AI as such. Mark the reasoning behind the decision.
- **Change of position:** Don't penalise a student for changing or keeping their Stage 1 position. Mark how well they explain the decision.

- **Worldview:** Don't mark a student's personal beliefs or worldview. Mark the evidence, reasoning and responsibility shown in their work.

A different question for higher education

If educational success is measured mainly by assignments and examinations, AI will inevitably challenge established assumptions about assessment and competence.

If the purpose of higher education is to develop people who can think independently, evaluate knowledge, recognise uncertainty and exercise responsible judgement, the challenge becomes more interesting. Universities must then ask whether their learning environments help students evaluate AI's answers, understand their implications and reach their own conclusions.

The challenge is not simply to educate students to use artificial intelligence. It is to develop students who, in an increasingly AI-mediated world, retain the capacity and willingness to think for themselves.

Transformation begins not with changing the world, but with changing the way we see it.

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